ECO 319/619

Time Series Analysis and Forecasting
Project Report

Topic - Effects of EV (Electric Vehicle) sales on Gasoline Prices in India

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Research Problem:

How the sales of EV (Electric Vehicles) affect the Gasoline Prices in India?

Introduction:

India is continuously moving towards achieving green energy goals. The recent development and adoption of electric vehicles has been a major progress towards achieving a green economy. As transportation is major contributor towards carbon emission, EV industry seems to be a good alternative in achieving decarbonization and dual carbon goals (Xiaolei Zhao., 2025, Xuemei Li.,2025, Anqi Wang.,2025, Jianjun Fang.,2025). India as a part of its Nationally Determined Contributions under the Paris Agreement is committed to reduce the emissions intensity of its GDP by 45% by 2030 from 2005 levels, and achieve net zero emissions by 2070 (IEEFA, 2025). As the transportation sector contributes to approximately 14% of India's energy sector carbon dioxide emission, necessary measures need to be taken in the sector of transportation. Many countries on the similar lines have set targets to increase the penetration of electric vehicles and decrease the dependence on fossil fuels by the year 2030 to reduce global warming and decrease temperature below 1.5-degree centigrade (Anika et al., 2022, Morgenthaler et al., 2021). Different policies like providing subsidies and financial incentives on purchase of electric vehicles are being formulated by countries across the globe (IEA, 2021a, Jin et al., 2022, Liu et al., 2022, Martins et al., 2023, Srivastava et al., 2022, Wangsa et al., 2023, Wappelhorst et al., 2020, Yang et al., 2022). These policies aim to decrease the dependence on fossil fuels driven vehicles and replace them with EVs ((Breschi et al., 2023, Glyniadakis and Balestieri, 2023, Hosein et al., 2023, Williams et al., 2023).

India is the third-largest carbon emitter and the second-largest importer of crude oil. The country has set an ambitious target to reduce its oil import dependence by replacing gasoline based vehicles with EV and producing more and more electricity from renewable and other clean energy sources to achieve a net zero target by 2070. India is expected to meet 50 % of its energy requirements from renewable energy by 2030 and decrease its excessive reliance on fossil fuels (Media Center, 2021). EV adoption decreases oil import bill thus increasing macroeconomic stability.

Literature Review:

Different studies have been conducted till now to know the perception of consumers towards EV vehicles. A study conducted by Singh et at.(2020) studies the elements that affect consumers' ability to adopt EV. Consumers are less

sensitive to change in electricity prices as compared to gasoline prices. ((Leard et al. (2021) and Gillingham et al. (2021)). According to Reuters, Tata Motors will be investing over US$ 2 billion in its electric vehicle business over the next five years, after raising funds from private equity firm TPG. In FY 25 e-2 wheeler had highest sales amongst all electric vehicles that was about 1,209,772 while e-rickshaw stands second highest with nearly 474,503 sales.(EVreporter, 2025). However policy support is essential to achieve the goal of 30% EV by 2030 in electric private cars.(IEEFA). This is likely due to pre conceived notion and attitude of people towards EV (Monika Jain and Archana Singh, 2024). According to e-AMRIT which is a portal especially dedicated towards electric vehicles and green mobility states that 1.32% of all vehicle sales in FY 2021-22 were electric. It also states that nearly 133% growth occurred in electric vehicles sales from FY 2015 to 2020. 380 EV manufacturers operate in India(e-AMRIT). Companies being competitive are also putting significant input in production of EVs. Mahindra Last Mile Mobility registered highest sales for e- 3W passenger and e-3W Cargo with 52,540 and 7.447 units respectively. Bajaj Auto was the highest gainer in both the categories.(EVreporter, 2025). According to the FY 24-25 report Uttar Pradesh ranks 1st in the list of states that have adopted the highest EVs with nearly 3,77,565 EVs being sold. (EVreporter, 2025). The transportation sector till now has been heavily dependent on two major petroleum products, namely Gasoline and high-speed diesel (HSD) (EIA, 2021). The more and more dependence on EV is expected to create downward pressure on oil price (Greene et al., 2018). The adoption of EVs highly depends on government incentives and policies. If there is large-scale adoption of EVs then the demand for oil could be decreased by 17% by 2030. (Chaturvedi, B.K., 2022). According to a study conducted by BloombergNEF in 2022, every 1 million EVs reduce the demand for oil by 50,000 barrels per day. EV price parity could decrease oil demand by 8 to 10 % by 2030 (McKinsey, 2020). Increase in cost of crude oil depreciates rupee and makes trade balance worse and raises inflation through cost push effects (Narayan and Narayan, 2020, Singh and Sharma, 2020). The long run demand for oil will be substituted if there is a gradual positive shift towards adoption of Evs (Chaturvedi, B.K., 2022).

The novelty of this study can be summarized as follows:

1. This study tries to analyze impact of recently emerging EV industry on traditional gasoline prices
2. The data used for above study is recent data and helps us in understanding the recent trends in EV industry
3. The study is exclusively carried out in India and analyzes EV sales and gasoline prices both from an Indian perspective.
4. The study also takes into account the effect of Russia Ukraine war on Indian gasoline prices and its effects on sales of EV in India.

Theoretical framework:

The theoretical framework used for above study includes major economic theories.

1. Supply Demand Theory: The fundamental theory of supply demand states that as the price of gasoline increases people will eventually shift towards alternative options which will increase the sales of electric vehicles.
2. Energy Substitution Theory: As EVs and gasoline are substitutes of each other in transportation, hence they can affect consumer preferences by their price elasticity of substitution mechanism. Thus when gasoline prices increase EV demand increases and vice versa.
3. Environmental economics theory: Consumers that adopt green mobility are awarded with subsidies while those polluting the environment are regulated with taxes and other regulations.
4. Behavioral economics theory: People don't choose EV just as a cost saving mechanism. Social influence and other people's perception towards electric vehicles also matter during the choice of buying an EV for oneself. This will continue to happen until EVs become the new normal in society.
5. Policy intervention theory: Government policies help in fast transition of technological adoptions

Data Description:

The data span ranges from January 2015 to June 2025. The data on sales of electric vehicles is taken from VAHAN SEWA portal of Indian government. The data mainly consists of electric battery operated vehicles(BOV) along with plug-in hybrid EV, Pure EV, strong hybrid EV. Gasoline prices in INR are taken from a site called Indexmundi. Exchange rate(LAV in %) is taken from FEDAI (Foreign Exchange Dealers' Association of India). CPI is taken from MOSPI (Ministry of Statistics and Programme Implementation). Brent crude oil price(INR) is taken from a website called Macrotrends. The reason why Brent crude oil is taken for this analysis is that it has more influence on international oil prices as compared to WTI. The time period covers important events and policy periods like FAME1 (Faster Adoption and Manufacturing of Hybrid and Electric Vehicles) that was launched by Indian government from April 1, 2015 to March 31, 2019 and FAME2 which had time period of April 1, 2019 to March 31, 2024. Along with this the data also covers covid-19 time period and tries to study effects of Russia Ukraine war which began in February, 2014 but got converted into full scale war in February, 2022. The variables in the data include YEAR, TOTAL_EV (representing total EV sales in a particular month), GASOLINE PRICES (INR), EXCHANGE RATE (LAV IN %), BRENT CRUDE OIL PRICE (INR), CPI (Consumer Price Index).

YEAR	TOTAL_EV	GASOLINE PRICES (INR)	EXCHANGE RATE (LAV IN %)
Length:126	Min. : 168	Min. : 45.18	Length:126
Class :character	1st Qu.: 6631	1st Qu.:102.64	Class :character
Mode :character	Median : 14089	Median :128.09	Mode :character
	Mean : 52123	Mean :146.68	
	3rd Qu.:102830	3rd Qu.:185.64	
	Max. :229673	Max. :318.72	

BRENT CRUDE OIL PRICE(INR)	CPI
Min. : 1318	Min. :119.5
1st Qu.: 4618	1st Qu.:134.0
Median : 5924	Median :150.7
Mean : 5909	Mean :154.6
3rd Qu.: 6903	3rd Qu.:175.7
Max. :11140	Max. :196.5

Table 1: Descriptive statistics

FAME1	FAME2	COVID	RUS_UKR_WAR
Min. :0.000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.000	Median :0.0000	Median :1.0000	Median :0.0000
Mean :0.381	Mean :0.4762	Mean :0.5079	Mean :0.3254
3rd Qu.:1.000	3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:1.0000
Max. :1.000	Max. :1.0000	Max. :1.0000	Max. :1.0000

Table 2: Summary Statistics

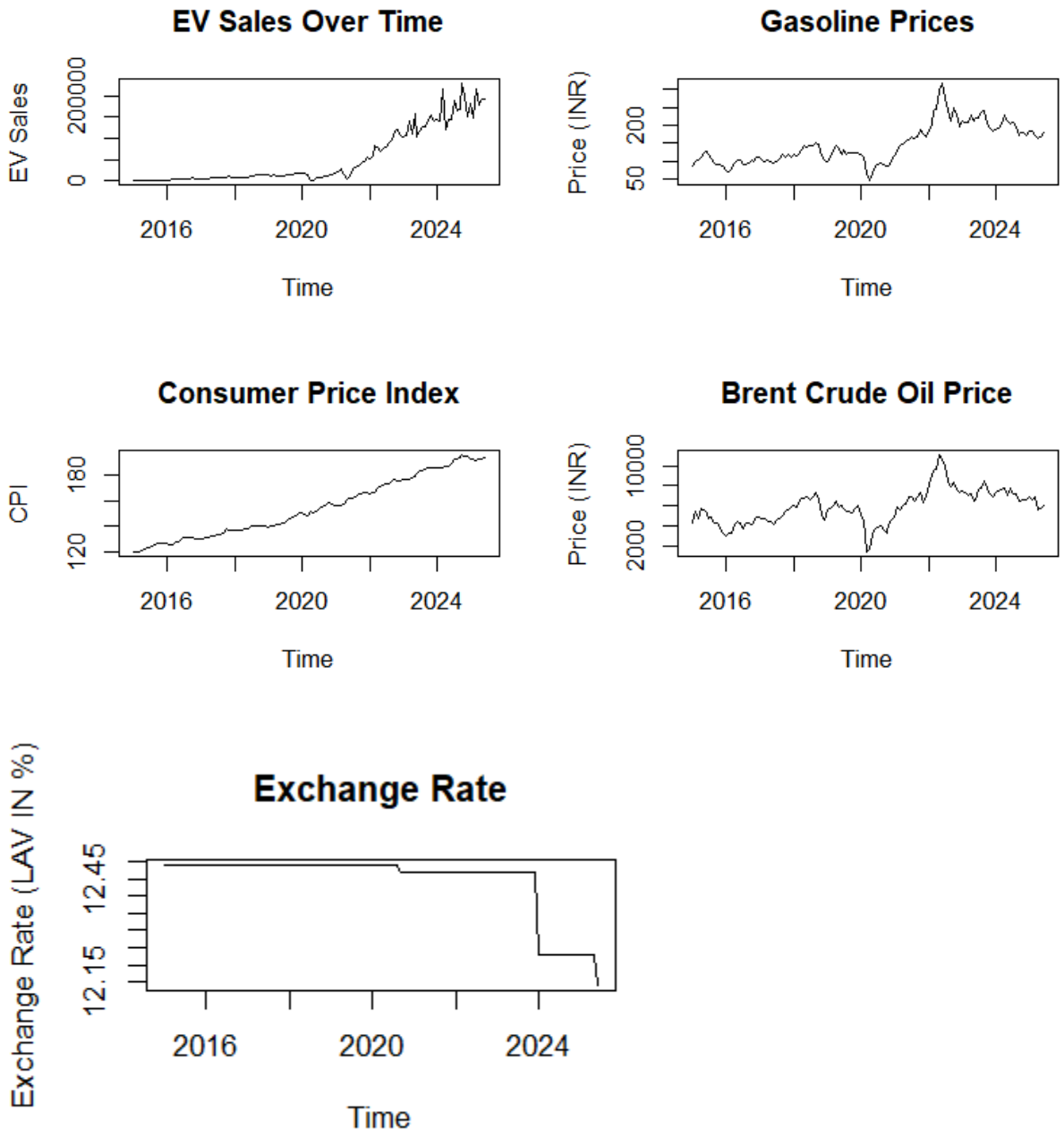


Table 3: Time series plot of key variables (2015-2025)

Methodology:

The research uses R Studio software to make the analysis and forecasting. As residuals are not independent over time, there exists a serial correlation in data. Also skewness shows significant asymmetry and kurtosis shows extreme values (heavy tails). Therefore we use Bayesian Vector Autoregression (BVAR) which treats model parameters as random with probability distribution instead of normal

VAR which treats parameters as fixed values.

Unit root test

We use the ADF test for checking unit root for time series variables where the null hypothesis represents the variable is non-stationary and has unit root while the alternate hypothesis represents that data is stationary.

Variables	p value	Results
TOTAL_EV	0.9578	Non stationary
GASOLINE PRICES	0.4442	Non stationary
EXCHANGE RATE	0.9781	Non stationary
BRENT CRUDE OIL PRICE	0.3925	Non stationary
CPI	0.5797	Non stationary

Table 4: Results of ADF test

The results of ADF suggest that the data is non-stationary and needs differencing. Therefore we difference the data and again perform the ADF test. On differencing we obtain p value equal to 0.01 for all variables, thus on differencing the data becomes stationary.

Model Estimation:

We use Bayesian VAR which helps us understand how each variable influences each other over time. The model includes 121 observations, 5 endogenous variables and 4 exogenous dummy variables. We choose 4 lags on the basis of a lag test in which 4 number of lags show good MCMC draws of 10,000. The equation for estimation of BVAR(4) is

$$\begin{bmatrix} \mathrm{TOTAL_EV}_t \\ \mathrm{GASOLINE_PRICE}_t \\ \mathrm{EXCHANGE_RATE}_t \\ \mathrm{BRENT_OIL}_t \\ \mathrm{CPI}_t \end{bmatrix} = A_0 + \sum_{i=1}^{4} A_i \begin{bmatrix} \mathrm{TOTAL_EV}_{t-i} \\ \mathrm{GASOLINE_PRICE}_{t-i} \\ \mathrm{EXCHANGE_RATE}_{t-i} \\ \mathrm{BRENT_OIL}_{t-i} \\ \mathrm{CPI}_{t-i} \end{bmatrix} + B_1 \mathrm{FAME1}_t +$$

$$B_2\mathrm{FAME2}_t + B_3\mathrm{COVID}_t + B_4\mathrm{UKRAINE_WAR}_t + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \\ \varepsilon_{3t} \\ \varepsilon_{4t} \\ \varepsilon_{5t} \end{bmatrix}$$

We then carry out Granger causality tests in both directions to check if electric vehicles affect gasoline prices or gasoline prices affect sales of electric vehicles or both. Later we perform an impulse response function. The IRF in BVAR checks the effects of shock of one endogenous variable on the other endogenous variables. We then look at forecast error variance decomposition (FEVD) of the model followed by forecasting the model.We use VAR model for making forecasting as results from VAR are more conclusive and helps us in better forecasting as compared to BVAR

Results:

	TOTAL_EV	GASOLINE PRICES (INR)	EXCHANGE RATE (LAV IN %)
TOTAL_EV	1.0000000	0.7116028	-0.8063883
GASOLINE PRICES (INR)	0.7116028	1.0000000	-0.3802560
EXCHANGE RATE (LAV IN %)	-0.8063883	-0.3802560	1.0000000
BRENT CRUDE OIL PRICE(INR)	0.5537372	0.9290346	-0.2776099
CPI	0.9072351	0.7631661	-0.7066925

	BRENT CRUDE OIL PRICE(INR)	CPI
TOTAL_EV	0.5537372	0.9072351
GASOLINE PRICES (INR)	0.9290346	0.7631661
EXCHANGE RATE (LAV IN %)	-0.2776099	-0.7066925
BRENT CRUDE OIL PRICE(INR)	1.0000000	0.6107497
CPI	0.6107497	1.0000000

Table 5: Correlation matrix

The correlation matrix between EV sales and CPI shows EV sales are highly linked to overall economic activity while that between EV sales and gasoline prices show higher gasoline prices boost EV sales.
The results from BVAR model analysis are as follows

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Bayesian VAR consisting of 121 observations, 5 variables and 4 lags.
Time spent calculating: 2.46 secs
Hyperparameters: lambda
Hyperparameter values after optimisation: 0.009
Iterations (burnt / thinning): 10000 (5000 / 1)
Accepted draws (rate): 771 (0.154)

Numeric array (dimensions 21, 5) of coefficient values from a BVAR.
Median values:
                                TOTAL_EV GASOLINE PRICES (INR) EXCHANGE RATE (LAV IN %)
constant                        3978.848                -0.175                       0
TOTAL_EV-lag1                     -0.735                 0.000                       0
GASOLINE PRICES (INR)-lag1        -8.187                 0.916                       0
EXCHANGE RATE (LAV IN %)-lag1     14.868                -0.002                       1
BRENT CRUDE OIL PRICE(INR)-lag1    0.823                -0.002                       0
CPI-lag1                          32.274                 0.008                       0
TOTAL_EV-lag2                     -0.412                 0.000                       0
GASOLINE PRICES (INR)-lag2        -3.894                -0.006                       0
EXCHANGE RATE (LAV IN %)-lag2     -8.010                 0.003                       0
BRENT CRUDE OIL PRICE(INR)-lag2    0.772                -0.006                       0
CPI-lag2                          -3.519                -0.008                       0
TOTAL_EV-lag3                     -0.310                 0.000                       0
GASOLINE PRICES (INR)-lag3        -1.195                 0.001                       0
EXCHANGE RATE (LAV IN %)-lag3      4.076                -0.005                       0
BRENT CRUDE OIL PRICE(INR)-lag3   -0.191                 0.000                       0
CPI-lag3                           0.808                -0.003                       0
TOTAL_EV-lag4                     -0.082                 0.000                       0
GASOLINE PRICES (INR)-lag4         0.568                 0.000                       0
EXCHANGE RATE (LAV IN %)-lag4     -3.179                 0.004                       0
BRENT CRUDE OIL PRICE(INR)-lag4    0.263                 0.000                       0
CPI-lag4                           1.823                -0.001                       0
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                                BRENT CRUDE OIL PRICE(INR)   CPI
constant                                             1.398 0.006
TOTAL_EV-lag1                                       -0.002 0.000
GASOLINE PRICES (INR)-lag1                          -0.661 0.000
EXCHANGE RATE (LAV IN %)-lag1                       -1.008 0.000
BRENT CRUDE OIL PRICE(INR)-lag1                      0.264 0.000
CPI-lag1                                             0.834 0.992
TOTAL_EV-lag2                                       -0.002 0.000
GASOLINE PRICES (INR)-lag2                          -0.078 0.000
EXCHANGE RATE (LAV IN %)-lag2                        0.152 0.000
BRENT CRUDE OIL PRICE(INR)-lag2                     -0.051 0.000
CPI-lag2                                            -0.172 0.000
TOTAL_EV-lag3                                        0.002 0.000
GASOLINE PRICES (INR)-lag3                          -0.036 0.000
EXCHANGE RATE (LAV IN %)-lag3                       -0.435 0.000
BRENT CRUDE OIL PRICE(INR)-lag3                     -0.044 0.000
CPI-lag3                                            -0.055 0.000
TOTAL_EV-lag4                                        0.002 0.000
GASOLINE PRICES (INR)-lag4                          -0.013 0.000
EXCHANGE RATE (LAV IN %)-lag4                        0.078 0.000
BRENT CRUDE OIL PRICE(INR)-lag4                     -0.016 0.000
CPI-lag4                                            -0.017 0.000

Numeric array (dimensions 5, 5) of variance-covariance values from a BVAR.
Median values:
                             TOTAL_EV GASOLINE PRICES (INR) EXCHANGE RATE (LAV IN %)
TOTAL_EV                 220474234.531             10112.664                  -16.438
GASOLINE PRICES (INR)        10112.664               286.512                    0.001
EXCHANGE RATE (LAV IN %)       -16.438                 0.001                    0.001
BRENT CRUDE OIL PRICE(INR)  -769865.673              5602.155                   -1.227
CPI                           -1901.965                -0.225                    0.000
                         BRENT CRUDE OIL PRICE(INR)       CPI
TOTAL_EV                           -769865.673 -1901.965
GASOLINE PRICES (INR)                 5602.155    -0.225
EXCHANGE RATE (LAV IN %)                -1.227     0.000
BRENT CRUDE OIL PRICE(INR)          391500.224   -22.430
CPI                                    -22.430     1.274

Log-Likelihood: -2707.359
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Table 6: Results of BVAR model

The hyperparameter sets the value of lambda at 0.009 automatically, iterations are equal to 10,000. For the total ev sales, the lags initially show negative value in its own equation then slowly shifting towards positive value. This might indicate mean reversion in sales after past spikes in EV sales. There is no impact EV sales on gasoline prices in the short run as indicated by zero lag values in the gasoline equation. EV sales also do not impact exchange rates. The small negative lag values in Brent crude oil price which later turn into positive value states that there is a possibility that EV sales might decrease the price of Brent crude oil in near future.

Looking at the gasoline prices lags, and its effect on EV sales, it is seen that the first three lags are negative indicating that as gasoline prices rise EV sales fall. However this effect is short lived and EV sales recover after some time as indicated by fourth positive lag. This also indicates the fact that consumers

initially might not be comfortable with moving towards changing the way their transportation works, however after considering the high fuel prices they might shift towards EVs gradually. High value in its own first lag indicates that gasoline prices are affected by its own past prices significantly. The prices beyond one month have little impact on determining the gasoline price. Exchange rate and CPI are not influenced much by changes in gasoline prices as observed by zero lag values. All negative values of lags in the Brent crude oil price equation show that gasoline prices take time to adjust with changes in Brent crude oil price hence there is a delay effect.

In case of exchange rate, the first lag which is highly positive indicates EV sales rises in the current month due to depreciation of rupee in previous month. Weaker rupee increases the import cost of oil which in turn motivates customers to shift towards electric vehicles. However, sales of EV due to rupee depreciation is not sustained. Third and fourth alternate sign lags show short term macroeconomic shocks.

The short run fluctuations in INR/USD do not significantly affect domestic fuel prices. This can be seen by tiny alternating lag values. The first lag of the exchange rate in its own equation is 1 which shows that the exchange rate is highly dependent on its own past values and beyond a month's statistics are not much useful in predicting future exchange rate. Short run changes in exchange rate do not affect CPI as observed by zero value of lags. No consistent causal relationship is seen between exchange rate and Brent crude oil price. This is shown by the weak rupee at lag one correlates with lower crude oil prices mainly due to reverse causality.

Alternating values of signs of Brent crude oil price in the TOTAL_EV equation shows that as gasoline prices rise, sales of electric vehicles also rise followed by some correction. In the gasoline equation the values of lags and tending towards zero suggest a weak and slow link between gasoline price and Brent crude oil price. The lags show that Brent crude oil prices do not affect the exchange rate and CPI in the short run while mean reversion is found in the case of its own equation.

No clear evidence of how CPI affects sales of EV is shown. It has kind of mixed effects that are economically small. High inflation a month ago might increase EV sales today but no clear cut evidence is shown. High inflation last month could increase gasoline prices this month as shown by the first positive lag. Other small and negative lag suggest mean reversion. No short term relationship with exchange rate is seen as indicated by zero values of lags. Unclear transmission channel from domestic inflation to global oil prices is seen as seen in case of Brent crude oil price. The first lag is highly positive in its own equation while all other lags are zero indicating CPI is mainly affected by its own past values in the short run.

The results from the variance covariance matrix shows TOTAL_EV and BRENT CRUDE OIL PRICE show highest residual volatility. The covariance

between EV sales and gasoline prices is 10,112.664 indicating that unexpected rise in gasoline prices can lead to unexpected increase in EV sales. This is likely due to an overall economic boost which increases prices and EV sales and gasoline prices. On the other hand, CPI and exchange rate show least variance. On the basis of correlation matrix, we can say that a 10% increase in gasoline prices can lead to 7% increase in EV sales.

Robustness Check:

For the robustness checks we will plot trace and density of ml and lambda

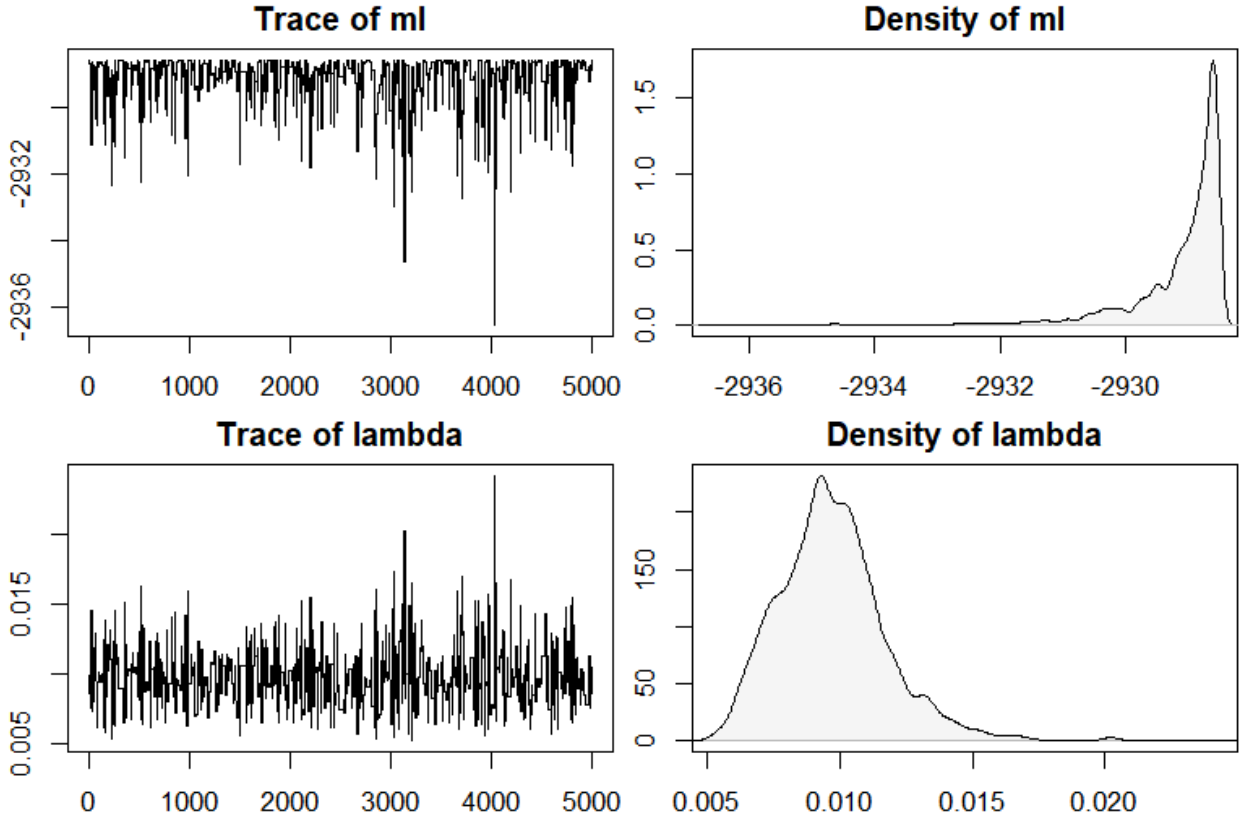


Table 7: Robustness Check

The effects of dummy variables as given as

	TOTAL_EV	GASOLINE PRICES (INR)	EXCHANGE RATE (LAV IN %)
FAME1	0.6970918	-0.0006374465	-1.812318e-06
FAME2	4.8978007	-0.0037921430	8.868100e-06
COVID	0.2372271	-0.0002512031	-2.201474e-06
RUS_UKR_WAR	0.3408273	-0.0011162594	-9.024258e-07

	BRENT CRUDE OIL PRICE(INR)	CPI
FAME1	-0.02023929	-7.201771e-05
FAME2	0.10104699	-2.032445e-04
COVID	-0.01462926	-7.795452e-07
RUS_UKR_WAR	-0.06380327	-2.480987e-04

Table 8: Effects of dummy variables

All the dummies show positive effects towards EV adoption while they do not affect gasoline prices much.

The results from Granger causality test between GASOLINE PRICES and TOTAL_EV shows p-value of 0.04238 which is statistically significant which tells us that GASOLINE PRICES Granger causes TOTAL_EV. The Granger causality test between TOTAL_EV and GASOLINE PRICES shows p-value of 0.4302 which is statistically insignificant indicating EV sales do not granger cause gasoline prices.

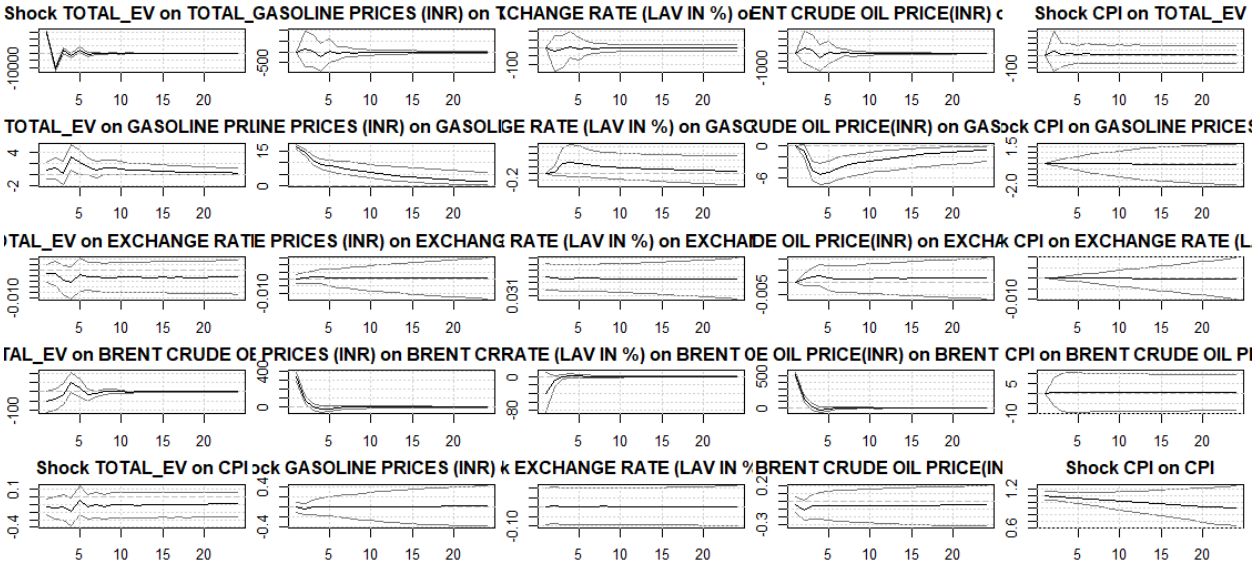


Table 9: Impulse Response Function

Each small graph in the IRF plot shows how one variable responds to shock given to another variable.The plots show us that sudden rise in EV sales decreases the gasoline prices however this effect is short lived and temporary.

EV sales do not have a consistent effect on exchange rate as seen by small responses. In the case of Brent crude oil, it is seen that higher EV adoption could decrease global crude oil prices. CPI is not affected by EV sales much. Therefore, we can say that EV sales affect gasoline prices however the effects are small and temporary.

In the IRF plots of Gasoline prices, we see higher costs of gasoline increase demand of EV however this effect is short lived. We see that as fuel price increases inflation also increases. No significant effect is seen in the case of Brent crude oil price and exchange rate. Weaker rupee makes import of crude oil costly as seen in case of gasoline and Brent crude oil prices and these higher import costs also add to inflation as seen in CPI. But we see EV adoption isn't affected much by currency movements. Increase in global crude oil prices increases domestic fuel prices. Brent crude oil prices show slow adoption of EV due to adjustment in consumer sentiments. Oil shocks do not affect currency much. In the case of CPI it is largely affected by other components and it does not have direct effect on other variables.

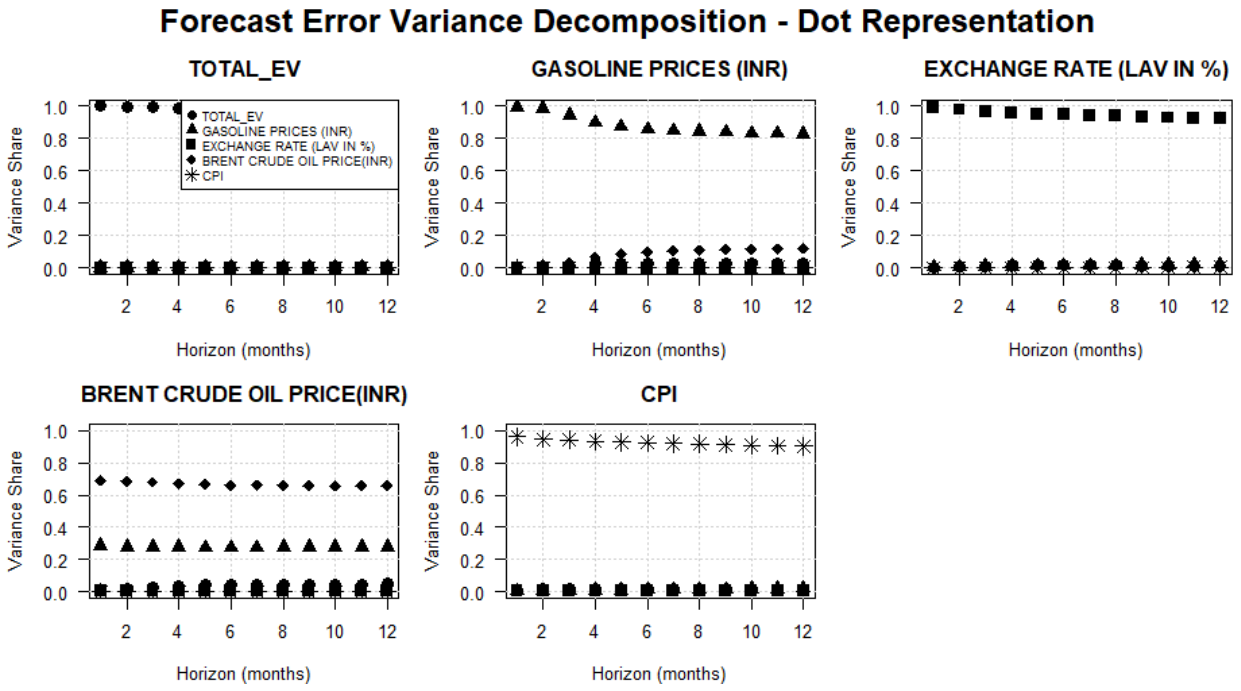


Table 10: Forecast Error Variance Decomposition

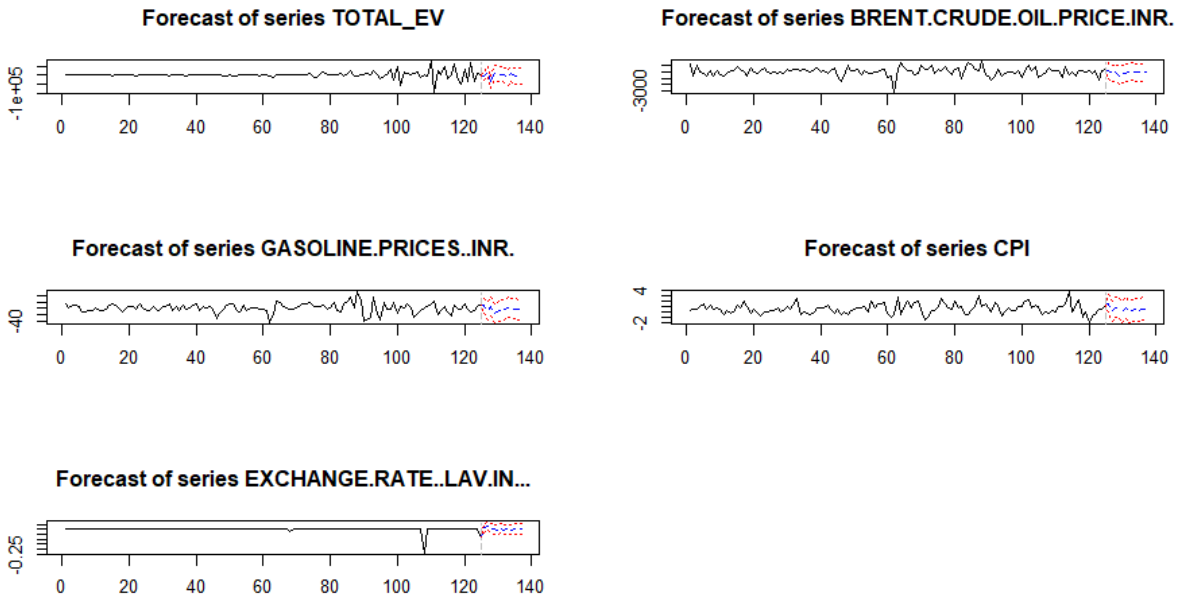


Table 11: Forecasting

The results of the forecast shows that EV sales are expected to remain stable, however there is a probability of steady growth in near future. Brent crude oil prices are expected to remain stable with mild fluctuations. Gasoline prices, CPI and exchange rates are expected to remain stable suggesting price stability and minimal volatility in case of exchange rate.

Conclusion:

The study finds that the electric vehicles industry right now is in its initial stages but an emerging market in future. The analysis from the BVAR model shows that sales of electric vehicles do not significantly affect the gasoline prices in India. Gasoline prices are mostly affected by international macro parameters like Brent crude oil price, exchange rate, CPI and also majorly by its own past price value. The results from granger causality test shows an interesting finding that sales of EV do not affect gasoline prices but gasoline prices do affect sales of EV. Hence, there exists a unidirectional relationship between gasoline prices and EV sales. The results also show that the overall economic condition of a nation also plays a crucial role in determining the sales of EV as well as gasoline prices.

The finding also suggests that FAME 1 and FAME 2 policies that were launched to increase the adoption of EV vehicles among consumers were largely successful, especially FAME 2. Both the FAME policies increased the sales of EV to great extent which is consistent with the policy intervention theory. However they had almost negligible impact on gasoline prices.

The covid-19 shows a slight positive effect likely due to the fact that the dummy

also captures the post recovery phase as well. The Russia-Ukraine war shows a positive effect on sales of EV, likely due to the fact that war increased the gasoline prices which made consumers shift towards electric vehicles adoption which is also consistent with the theory of energy substitution.

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